

QUIZ SURVIVAL GUIDE

Overdetermined & Homogeneous Linear Systems, Gradients, Circle Fitting

Source-grounded preparation from your two uploaded course PDFs

How to use this tonight: Read the Emergency Formula Sheet first. Then study the VERY HIGH cards, followed by HIGH cards. Use the ranked distractors to train recognition of traps; the correct answer is intentionally listed first in this study guide, unlike a real quiz.

Likelihood note: The ordering below is an inference from what the source PDFs emphasize and from the stated quiz topics. It is not a guarantee of what your instructor will ask.

Source-fidelity note: Most content below is directly supported by the uploaded notes. Added background is explicitly labeled. Two apparent source typos are flagged near the end so you do not memorize them as mathematics.

Study map

1. Emergency formula sheet - memorize these first
2. VERY HIGH probability question archetypes
3. HIGH probability trap questions
4. MEDIUM probability / spillover from the same PDFs
5. Trick-wording decoder and common distractors
6. Numerical micro-drills
7. Four 5-question mini-mocks (official-size simulations)
8. Source notes, typos, and final 10-minute checklist

1. Emergency formula sheet

If you memorize only one page: Know the formulas and the meaning of each symbol below. Most multiple-choice traps are sign, transpose, symmetry, or left-vs-right-side mistakes.

Concept	Must-know formula / fact	Source
Gradient definition	$\text{grad}_x J = [dJ/dx_1, dJ/dx_2, \dots, dJ/dx_N]^T$	Matrix p.2
Squared L2 norm	$\ v\ _2^2 = v^T v$	Matrix pp.2,4
Derivative of $x^T x$	$\text{grad}_x (x^T x) = 2x$	Matrix p.2
Derivative of $x^T a$	$\text{grad}_x (x^T a) = a$	Matrix pp.2-3
Quadratic form	$\text{grad}_x (x^T A x) = (A^T + A)x$	Matrix p.3
Symmetric quadratic form	If $B^T = B$: $\text{grad}_x (x^T B x) = 2Bx$	Matrix p.3
Least-squares objective	$x^* = \arg \min_x \ Ax - b\ _2^2$	Matrix pp.3-4
Least-squares gradient	$\text{grad} J = 2A^T(Ax - b) = 2A^T A x - 2A^T b$	Matrix p.4
Normal equation	$A^T A x = A^T b$	Matrix p.4
Closed form (when invertible)	$x = (A^T A)^{-1} A^T b$	Matrix p.4
Circle	$(x-x_0)^2 + (y-y_0)^2 = R^2$	Circle/Ellipse p.1
Circle linear row	$[1, -2x_i, -2y_i] [c, x_0, y_0]^T = -x_i^2 - y_i^2$	Circle/Ellipse p.1
Circle auxiliary parameter	$c = x_0^2 + y_0^2 - R^2$	Circle/Ellipse p.1
Recover radius	$R = \sqrt{x_0^2 + y_0^2 - c}$	Derived directly from source definition of c
Circle normal	$\text{grad} J = [2(x-x_0), 2(y-y_0)]^T$	Circle/Ellipse p.3
Ellipse/conic equation	$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$	Circle/Ellipse p.2
Ellipse condition	$B^2 - 4AC < 0$	Circle/Ellipse p.2
Ellipse homogeneous fit	$Mz = 0, z = [A, B, C, D, E, F]^T$	Circle/Ellipse p.2
Ellipse scale constraint	$B^2 - 4AC = -1 \Leftrightarrow z^T K z = -1$	Circle/Ellipse pp.2-3
Generalized eigenproblem	$M^T M z = -\lambda K z$	Circle/Ellipse p.3

One-line mental model: Inhomogeneous $Ax=b \rightarrow$ minimize residual if overdetermined. Homogeneous $Mz=0 \rightarrow$ zero is always available, so fitting needs a nonzero scale constraint. Circle fitting is linear after regrouping unknown parameters.

2. VERY HIGH probability question archetypes

1. Recognize an overdetermined system [VERY HIGH]

Core / likely wording: If A is $m \times n$, when is $Ax=b$ overdetermined?

Tricky formulations: What dimension inequality means more equations than unknowns? | When does the source call $Ax=b$ an estimation problem? | Which relation between rows and columns signals least squares?

Source basis: Matrix p.3.

Ranked answer possibilities (best first):

1. $m > n$ (more equations than unknowns)
2. $m = n$ (square system - common trap)
3. $m < n$ (underdetermined)
4. A must be symmetric (irrelevant)

Why correct: The Matrix notes state that if A is not square and $m > n$, the problem is an estimation problem and the mismatch should be minimized.

Memorize / understand: m = equations (rows), n = unknowns (columns). OVERdetermined means too many equations: $m > n$.

2. Least-squares objective [VERY HIGH]

Core / likely wording: What quantity is minimized for an overdetermined inhomogeneous system $Ax=b$?

Tricky formulations: Which residual is minimized? | What is the standard least-squares cost? | Which arg min corresponds to the normal equation?

Source basis: Matrix pp.3-4.

Ranked answer possibilities (best first):

1. $\|Ax - b\|_2^2$
2. $\|A^T x - b\|_2^2$ (transpose in wrong place)
3. $\|Ax\|_2^2$ (forgets b)
4. $\|x - b\|_2^2$ (dimension/meaning trap)

Why correct: The source defines the standard least-squares estimate as $\arg \min_x \|Ax - b\|_2^2$.

Memorize / understand: Residual = prediction minus target = $Ax - b$. Least squares literally squares the length of that residual.

3. Least-squares gradient [VERY HIGH]

Core / likely wording: For $J = \|Ax - b\|_2^2$, what is $\text{grad}_x J$?

Tricky formulations: Differentiate the squared residual norm. | Which gradient leads to the normal equation? | Which expression must be set to zero at the optimum?

Source basis: Matrix p.4.

Ranked answer possibilities (best first):

1. $2A^T(Ax - b) = 2A^T A x - 2A^T b$
2. $2(Ax - b)$ (missing chain-rule A^T)
3. $2A(Ax - b)$ (A instead of A^T)
4. $(A^T + A)x$ (wrong formula: quadratic-form rule without b)

Why correct: Expanding and differentiating gives $2A^T A x - 2A^T b$. Setting it to zero produces the normal equation.

Memorize / understand: Squared residual + chain rule: outside gives 2, inside Ax gives A^T on the left.

4. Normal equation [VERY HIGH]

Core / likely wording: Which equation characterizes the least-squares optimum?

Tricky formulations: After setting the gradient to zero, what equation remains? | Which is called the normal equation? | Which equation does the least-squares solution satisfy?

Source basis: Matrix p.4.

Ranked answer possibilities (best first):

1. $A^T A x = A^T b$
2. $A A^T x = b$ (wrong multiplication order)
3. $Ax = A^T b$ (missing A^T on left)
4. $A^T x = b$ (dimension/transpose trap)

Why correct: The source explicitly names $A^T A x = A^T b$ as the normal equation.

Memorize / understand: Transpose both sides of the residual geometry mentally: $A^T(Ax-b)=0$.

5. Closed-form least-squares solution [VERY HIGH]

Core / likely wording: Assuming $A^T A$ is invertible, what is the standard solution?

Tricky formulations: Which formula is the left pseudoinverse form? | Solve the normal equation for x . | Which matrix is inverted?

Source basis: Matrix p.4.

Ranked answer possibilities (best first):

1. $x = (A^T A)^{-1} A^T b$
2. $x = A^{-1} b$ (A may be non-square)
3. $x = A^T (A A^T)^{-1} b$ (different rank/shape context)
4. $x = (A A^T)^{-1} A b$ (wrong order/dimensions)

Why correct: Left-multiplying the normal equation by $(A^T A)^{-1}$ yields the stated solution.

Memorize / understand: For tall A , remember TALL $\rightarrow$ transpose first: $A^T A$ becomes $n \times n$ and can be inverted if full column rank.

6. Gradient of $x^T x$ [VERY HIGH]

Core / likely wording: What is $\text{grad}_x (x^T x)$?

Tricky formulations: Derivative of squared L2 norm? | If $J = \sum_i x_i^2$, what is the vector derivative? | What is the special identity used repeatedly in the notes?

Source basis: Matrix p.2.

Ranked answer possibilities (best first):

1. $2x$
2. x (forgets derivative of square)
3. 2 (scalar instead of vector)
4. x^T (wrong orientation)

Why correct: $x^T x$ is the sum of squared components, so each partial derivative is $2x_i$.

Memorize / understand: Square $\rightarrow$ times 2. Vector of componentwise $2x_i$ values is simply $2x$.

7. Gradient of $x^T A x$ (general A) [VERY HIGH]

Core / likely wording: If A is not assumed symmetric, what is $\text{grad}_x (x^T A x)$?

Tricky formulations: Which formula is valid without symmetry? | What changes if A is not symmetric? | Choose between $2Ax$ and $(A^T + A)x$.

Source basis: Matrix p.3.

Ranked answer possibilities (best first):

1. $(A^T + A)x$
2. $2Ax$ (only guaranteed when A is symmetric)
3. $A^T A x$ (least-squares-looking trap)
4. Ax (missing the second contribution)

Why correct: The source derives one contribution from each occurrence of x , yielding $(A^T + A)x$.

Memorize / understand: x appears twice: one contribution gives A , the other gives A^T .

8. Symmetric quadratic-form shortcut [VERY HIGH]

Core / likely wording: If $B^T = B$, simplify $\text{grad}_x (x^T B x)$.

Tricky formulations: What does symmetry do to $(B^T + B)x$? | Special case for a symmetric matrix? | When is $2Bx$ valid?

Source basis: Matrix p.3.

Ranked answer possibilities (best first):

1. $2Bx$
2. $(B^T + B)x$ (correct but unsimplified; if asked simplified, $2Bx$ is best)
3. Bx (missing factor 2)
4. $2B^T Bx$ (wrong formula)

Why correct: For symmetric B , $B^T + B = B + B = 2B$.

Memorize / understand: Symmetric means transpose changes nothing, so the two terms become twins: $2B$.

9. Circle equation and expansion [VERY HIGH]

Core / likely wording: Which expanded expression is equivalent to $(x-x_0)^2 + (y-y_0)^2 = R^2$?

Tricky formulations: Expand the implicit circle equation. | Where do the $-2x_0x$ and $-2y_0y$ terms come from? | Which signs are correct after moving R^2 to the left?

Source basis: Circle/Ellipse p.1.

Ranked answer possibilities (best first):

1. $x^2 + y^2 - 2x_0x - 2y_0y + x_0^2 + y_0^2 - R^2 = 0$
2. $x^2 + y^2 + 2x_0x + 2y_0y + x_0^2 + y_0^2 - R^2 = 0$ (sign trap)
3. $x^2 + y^2 - x_0x - y_0y - R^2 = 0$ (missing factors 2)
4. $x^2 + y^2 - 2x_0x - 2y_0y - R^2 = 0$ (missing center-square terms)

Why correct: Apply $(u-v)^2 = u^2 - 2uv + v^2$ to both coordinates, then subtract R^2 .

Memorize / understand: Each squared difference contributes three terms: point-square, -2 cross term, center-square.

10. Circle-fitting design row [VERY HIGH]

Core / likely wording: For data point (x_i, y_i) , which row multiplies $[c, x_0, y_0]^T$?

Tricky formulations: Which coefficients belong on the left? | Choose the circle-fitting row from four plausible polynomial rows. | Where do the -2 factors belong?

Source basis: Circle/Ellipse p.1.

Ranked answer possibilities (best first):

1. $[1, -2x_i, -2y_i]$
2. $[x_i^2, y_i^2, 1]$ (plausible conic trap)
3. $[1, x_i, y_i]$ (missing -2)
4. $[-x_i^2 - y_i^2, x_i, y_i]$ (mixes RHS into matrix)

Why correct: After expansion and regrouping unknowns, c has coefficient 1, x_0 has coefficient $-2x_i$, and y_0 has coefficient $-2y_i$.

Memorize / understand: Circle row = ONE, minus TWO x , minus TWO y .

11. Circle-fitting unknown vector [VERY HIGH]

Core / likely wording: What unknown vector makes circle fitting linear in the notes?

Tricky formulations: Which parameter combination replaces direct R ? | Why is R not used directly as a linear parameter? | Identify the three unknowns in the matrix equation.

Source basis: Circle/Ellipse p.1.

Ranked answer possibilities (best first):

1. $[x_0^2 + y_0^2 - R^2, x_0, y_0]^T$
2. $[x_0, y_0, R]^T$ (natural but not linear after expansion)
3. $[x_0^2, y_0^2, R^2]^T$ (cross structure is wrong)
4. $[R^2, x_0 y_0, x_0 + y_0]^T$ (not from derivation)

Why correct: The constant terms $x_0^2 + y_0^2 - R^2$ are grouped into one auxiliary scalar so the remaining equation is linear in the fitted parameter vector.

Memorize / understand: Fit c, x_0, y_0 first; recover R afterward. Here c = center-norm-squared minus radius-squared.

12. Circle-fitting right-hand side [VERY HIGH]

Core / likely wording: For point (x_i, y_i) , what is the right-hand side of the source linear system?

Tricky formulations: Which squared point terms are moved to the right? | Is the RHS positive or negative? | Which option confuses left-side -2 coefficients with the RHS?

Source basis: Circle/Ellipse p.1.

Ranked answer possibilities (best first):

1. $-x_i^2 - y_i^2$
2. $x_i^2 + y_i^2$ (sign trap)
3. $-2x_i - 2y_i$ (these are left-side coefficients)
4. $R^2 - x_i^2 - y_i^2$ (R is not a known RHS quantity)

Why correct: The known point-square terms $x_i^2 + y_i^2$ are moved from the left side to the right, changing their sign.

Memorize / understand: Left: 1, $-2x$, $-2y$. Right: negative squares.

13. Circle fitting as an overdetermined linear problem [VERY HIGH]

Core / likely wording: How does the source solve circle fitting when many points are available?

Tricky formulations: Inverse or pseudoinverse? | What changes in the overdetermined case? | Which matrix operation is named explicitly?

Source basis: Circle/Ellipse p.1.

Ranked answer possibilities (best first):

1. Use the inverse when applicable, or the pseudoinverse in the overdetermined case
2. Use only a direct 3x3 inverse regardless of point count
3. Use a homogeneous eigenproblem (that is the ellipse section)
4. Differentiate with respect to R directly and ignore linearization

Why correct: The circle-fitting notes explicitly call the formulation an inhomogeneous linear problem and say to use the inverse or pseudoinverse, especially for the overdetermined case.

Memorize / understand: Circle = inhomogeneous linear least squares after regrouping. Ellipse = homogeneous constrained fit in these notes.

14. Recover the radius [VERY HIGH]

Core / likely wording: If $c = x_0^2 + y_0^2 - R^2$ is fitted, how do you recover R?

Tricky formulations: Solve the auxiliary parameter for R. | Which sign appears before c under the square root? | What if c is negative?

Source basis: Direct algebra from the circle parameter definition on Circle/Ellipse p.1.

Ranked answer possibilities (best first):

1. $R = \sqrt{x_0^2 + y_0^2 - c}$
2. $R = \sqrt{c - x_0^2 - y_0^2}$ (sign reversal)
3. $R = x_0^2 + y_0^2 - c$ (forgets square root)
4. $R = \sqrt{x_0 + y_0 - c}$ (forgets squares)

Why correct: Rearranging $c = x_0^2 + y_0^2 - R^2$ gives $R^2 = x_0^2 + y_0^2 - c$, and radius is the nonnegative square root.

Memorize / understand: Move c to the other side: radius-squared = center-squared-sum minus c.

15. Circle gradient / normal [VERY HIGH]

Core / likely wording: For $J = (x - x_0)^2 + (y - y_0)^2 - R^2$, what is $\text{grad } J$?

Tricky formulations: What is the normal to the circle? | Which vector points radially outward? | Does R appear in the gradient with respect to x,y?

Source basis: Circle/Ellipse p.3.

Ranked answer possibilities (best first):

1. $[2(x - x_0), 2(y - y_0)]^T$
2. $[x - x_0, y - y_0]^T$ (same direction but not the exact gradient)
3. $[2x_0, 2y_0]^T$ (differentiates wrong terms)
4. $[2(x - x_0), -2(y - y_0)]^T$ (sign trap)

Why correct: Differentiate J with respect to x and y; the constant $-R^2$ disappears. The source identifies this gradient as the curve normal.

Memorize / understand: Gradient of squared distance from center is twice the displacement from center.

3. HIGH probability trap questions

1. Gradient of a scalar product [HIGH]

Core / likely wording: If a is constant, what is $\text{grad}_x (x^T a)$ or $\text{grad}_x (a^T x)$?

Tricky formulations: Derivative of a linear scalar function? | Which vector survives differentiation? | Does transposing the scalar product change the derivative?

Source basis: Matrix pp.2-3.

Ranked answer possibilities (best first):

1. a
2. x (differentiates the wrong vector)
3. $2a$ (confuses with squared norm)
4. a^T (wrong orientation for the source convention)

Why correct: The source expands $x^T a = \sum_i a_i x_i$, giving partial derivative a_i for each component.

Memorize / understand: Linear in x means the coefficient vector is the gradient.

2. Why a gradient is normal to an implicit curve [HIGH]

Core / likely wording: Why does $\text{grad } J$ give a normal direction to the curve $J(x,y)=0$?

Tricky formulations: Is the gradient tangent or normal? | What is perpendicular to a level set? | Why does moving along the curve give zero directional change?

Source basis: The source states that curve normals are given by gradients (Circle/Ellipse p.3); geometric explanation is added background.

Ranked answer possibilities (best first):

1. The gradient is perpendicular to level sets of J , so it is normal to $J=0$
2. The gradient is always tangent to the curve
3. The gradient always has unit length
4. The gradient always points to the origin

Why correct: Along a level curve the function value stays constant, so tangent motion has zero first-order change. The gradient is therefore perpendicular to tangent directions.

Memorize / understand: Level set = same value. Gradient = steepest change, so it points across the level set, not along it.

3. Homogeneous system: trivial solution [HIGH]

Core / likely wording: For $Mz=0$, which solution always exists?

Tricky formulations: What is the unavoidable solution of any homogeneous linear system? | Why can unconstrained minimization of $\|Mz\|$ pick a useless answer? | Which vector automatically makes the residual zero?

Source basis: Ellipse section uses $Mz=0$ (Circle/Ellipse p.2); trivial-solution statement is standard background implicit in that setup.

Ranked answer possibilities (best first):

1. $z=0$
2. $z=1$ (not meaningful for a vector system)
3. $z=M^{-1}0$ only if M is square/invertible - unnecessary
4. No solution exists

Why correct: Multiplying any matrix by the zero vector gives the zero vector.

Memorize / understand: Homogeneous means zero on the right, so zero on the left is automatic with $z=0$.

4. Why homogeneous fitting needs a scale constraint [HIGH]

Core / likely wording: Why not simply minimize $\|Mz\|$ over all z ?

Tricky formulations: Why impose $B^2 - 4AC = -1$? | What ambiguity exists in homogeneous parameters? | What prevents the all-zero conic?

Source basis: Circle/Ellipse pp.2-3.

Ranked answer possibilities (best first):

1. Without a constraint, $z=0$ trivially minimizes the residual; scale is also arbitrary
2. Because M must be made square
3. Because homogeneous systems have no solutions
4. Because norms can become negative

Why correct: The ellipse notes state that parameter scale is arbitrary and fix it using $B^2 - 4AC = -1$ (equivalently $z^T K z = -1$). This excludes the zero vector and fixes scale.

Memorize / understand: Homogeneous fit has no natural magnitude: z and αz describe the same equation. Constrain the scale to avoid zero.

5. Ellipse/conic implicit form [HIGH]

Core / likely wording: What is the general quadratic form used for ellipse fitting?

Tricky formulations: Which terms appear in a general 2D quadratic curve? | Which coefficient multiplies xy ? | How many parameters are in z ?

Source basis: Circle/Ellipse p.2.

Ranked answer possibilities (best first):

1. $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$
2. $Ax^2 + By^2 + Cx + Dy + E = 0$ (missing xy and one parameter)
3. $Ax + By + C = 0$ (line, not conic)
4. $A(x-x_0)^2 + C(y-y_0)^2 = R^2$ (specialized form, not general conic)

Why correct: The source defines the general implicit quadratic curve with six coefficients A through F .

Memorize / understand: Quadratic in 2D: x^2 , xy , y^2 , x , y , constant - six terms.

6. Ellipse discriminant condition [HIGH]

Core / likely wording: According to the notes, when does the general quadratic represent an ellipse?

Tricky formulations: Sign of $B^2 - 4AC$? | Which discriminant inequality selects ellipses? | What distinguishes ellipse from parabola/hyperbola in this criterion?

Source basis: Circle/Ellipse p.2.

Ranked answer possibilities (best first):

1. $B^2 - 4AC < 0$
2. $B^2 - 4AC > 0$ (hyperbola-style trap)
3. $B^2 - 4AC = 0$ (parabola-style trap)
4. $A+C=0$ (not the stated criterion)

Why correct: The source explicitly states the ellipse condition $B^2 - 4AC < 0$.

Memorize / understand: Ellipse: discriminant negative. In the source, the scale-fixed version is set to -1 .

7. Ellipse design-matrix row [HIGH]

Core / likely wording: What row of M corresponds to a point (x_i, y_i) in the homogeneous conic fit?

Tricky formulations: Which polynomial features multiply $[A, B, C, D, E, F]$? | Do x^2 and y^2 appear before linear terms? | Which row has exactly six entries?

Source basis: Circle/Ellipse p.2.

Ranked answer possibilities (best first):

1. $[x_i^2, x_i y_i, y_i^2, x_i, y_i, 1]$
2. $[1, -2x_i, -2y_i]$ (circle-fit row)
3. $[x_i, y_i, 1, x_i^2, y_i^2, x_i y_i]$ (same ingredients, wrong coefficient order)
4. $[x_i^2, y_i^2, x_i, y_i, 1]$ (missing xy term)

Why correct: Each row contains the coefficients multiplying A, B, C, D, E, F in the general conic equation.

Memorize / understand: Match the parameter order $A B C D E F$ with $x^2, xy, y^2, x, y, 1$.

8. Ellipse parameter vector and homogeneous equation [HIGH]

Core / likely wording: Which compact system is used for ellipse fitting?

Tricky formulations: What is z ? | Is the right-hand side b or zero? | How is the polynomial equation stacked across points?

Source basis: Circle/Ellipse p.2.

Ranked answer possibilities (best first):

1. $Mz=0$ with $z=[A, B, C, D, E, F]^T$
2. $Mz=b$ with $z=[x_0, y_0, R]^T$
3. $Az=b$ with three unknowns
4. $M^T z=0$ with $z=[x, y]^T$

Why correct: The source stacks the general conic equation for all data points into the homogeneous system $Mz=0$.

Memorize / understand: Ellipse section: six coefficients, homogeneous zero RHS.

9. Ellipse constraint in matrix form [HIGH]

Core / likely wording: How is the scale/ellipse constraint represented compactly?

Tricky formulations: Equivalent of $B^2-4AC=-1$? | Which quadratic constraint is used in the Lagrangian? | What role does K play?

Source basis: Circle/Ellipse pp.2-3.

Ranked answer possibilities (best first):

1. $z^T K z = -1$
2. $z^T z = 0$ (forces $z=0$)
3. $Kz=0$ (different constraint)
4. $Mz=-1$ (confuses data matrix and constraint)

Why correct: The notes define a specific matrix K so that $z^T K z$ equals B^2-4AC , then fix it to -1 .

Memorize / understand: K encodes the conic-type/scale condition; M encodes the data points.

10. Ellipse gradient / normal (mathematically correct) [HIGH]

Core / likely wording: For $J = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$, what is the mathematically correct gradient?

Tricky formulations: Differentiate the conic equation with respect to x and y . | Which component contains $2Cy$? | Where does the Bxy term contribute?

Source basis: Circle/Ellipse p.4, with an apparent printed typo in the second component.

Ranked answer possibilities (best first):

1. $[2Ax + By + D, Bx + 2Cy + E]^T$
2. $[2Ax + By + D, 2Cx + Bx + E]^T$ (this is what the source PDF appears to print; likely typo)
3. $[Ax + By + D, Bx + Cy + E]^T$ (missing factors 2)
4. $[2Ax + 2By + D, 2Bx + 2Cy + E]^T$ (over-differentiates xy)

Why correct: d/dx of Bxy is By and d/dy is Bx ; d/dy of Cy^2 is $2Cy$. The source page appears to contain a typo in the second component, flagged later in this guide.

Memorize / understand: Differentiate one variable at a time: x -component sees $2Ax + By + D$; y -component sees $Bx + 2Cy + E$.

11. Pseudoinverse connection [HIGH]

Core / likely wording: For a tall full-column-rank A , which expression equals the least-squares pseudoinverse action?

Tricky formulations: How does A^+ relate to the normal-equation formula? | What matrix multiplies b ? | Which expression is the left pseudoinverse?

Source basis: Matrix p.4 + Circle/Ellipse p.1; equivalence stated here as standard background.

Ranked answer possibilities (best first):

1. $A^+ b = (A^T A)^{-1} A^T b$
2. $A^T (A A^T)^{-1} b$ (right-pseudoinverse form for different shape/rank conditions)
3. $A^{-1} b$ (A need not be square)
4. $(A^T A) A^T b$ (missing inverse)

Why correct: The Matrix notes give the normal-equation closed form, while the circle notes refer to the pseudoinverse in the overdetermined case. For full column rank, these expressions coincide.

Memorize / understand: Tall $A \rightarrow$ left pseudoinverse $\rightarrow (A^T A)^{-1} A^T$.

4. MEDIUM probability / spillover

1. Lagrangian stationarity for ellipse fitting [MEDIUM]

Core / likely wording: What stationarity equation follows from the ellipse-fitting Lagrangian?

Tricky formulations: Differentiate $z^T M^T M z + \lambda(z^T K z + 1)$. | Which terms get factor 2? | What equation leads to a generalized eigenproblem?

Source basis: Circle/Ellipse p.3.

Ranked answer possibilities (best first):

1. $2M^T M z + 2\lambda K z = 0$
2. $M z + \lambda K = 0$ (wrong dimensions)
3. $2Mz + \lambda z = 0$ (missing transposes/K)
4. $M^T M z = 0$ only (drops constraint)

Why correct: Both quadratic forms differentiate to twice their symmetric matrix times z , giving the source stationarity equation.

Memorize / understand: Quadratic form with symmetric S : derivative is $2Sz$. Here S is $M^T M$ or K .

2. Generalized eigenvalue equation [MEDIUM]

Core / likely wording: Which generalized eigenvalue problem appears in ellipse fitting?

Tricky formulations: Rearrange the Lagrangian stationarity condition. | What plays the two matrix roles in the generalized eigenproblem? | Where does $-\lambda$ appear?

Source basis: Circle/Ellipse p.3.

Ranked answer possibilities (best first):

1. $M^T M z = -\lambda K z$
2. $Mz = \lambda z$ (ordinary eigenproblem, wrong matrix)
3. $Kz = Mz$ (missing transpose/product)
4. $M^T M = -\lambda K$ (drops eigenvector z)

Why correct: Moving the constraint term to the other side of $2M^T M z + 2\lambda K z = 0$ yields the stated equation.

Memorize / understand: Data energy matrix $M^T M$ on the left; constraint matrix K on the right.

3. Choose among ellipse-fit candidates [MEDIUM]

Core / likely wording: How does the source choose the optimal candidate z after the generalized eigenproblem?

Tricky formulations: What quantity is minimized among candidates? | Up to how many candidate solutions are mentioned? | Which candidate is best?

Source basis: Circle/Ellipse p.3.

Ranked answer possibilities (best first):

1. Choose the candidate minimizing $z^T M^T M z$ (equivalently the squared residual energy)
2. Choose the largest $\|z\|$ (scale is constrained, not the objective)
3. Choose the largest eigenvalue automatically
4. Choose the candidate with largest F coefficient

Why correct: The source says at most six candidate solutions are obtained and the optimal one minimizes $z^T M^T M z$.

Memorize / understand: Candidate generation comes from the eigenproblem; final choice returns to the original fit error.

4. Full column rank and the homogeneous null space [MEDIUM]

Core / likely wording: If M has full column rank, what are the exact solutions of $Mz=0$?

Tricky formulations: What does full column rank imply about the null space? | Can a nonzero z satisfy $Mz=0$ when all columns are independent? | Which solution remains?

Source basis: Added linear-algebra background; "full column rank" is not explicitly defined in the uploaded PDFs.

Ranked answer possibilities (best first):

1. Only $z=0$
2. Infinitely many nonzero solutions (requires a nontrivial null space)
3. No solution at all (zero always works)
4. $z=M^{(-1)}0$ must be computed (M may be non-square)

Why correct: Full column rank means the columns are linearly independent, so the null space contains only the zero vector.

Memorize / understand: Independent columns $\rightarrow$ no nontrivial combination can cancel to zero.

5. Sphere fitting analogy [MEDIUM]

Core / likely wording: How does sphere fitting extend circle fitting?

Tricky formulations: What extra coefficient appears in each design row? | What is the RHS? | How many fitted center coordinates appear?

Source basis: Circle/Ellipse pp.1-2. The source also appears to contain a minor coordinate typo in the displayed sphere equation.

Ranked answer possibilities (best first):

1. Use row $[1, -2x_i, -2y_i, -2z_i]$, RHS $-x_i^2 - y_i^2 - z_i^2$, and center (x_0, y_0, z_0)
2. Use the same 3-column circle row and ignore z
3. Use a homogeneous six-parameter conic
4. Use $[x_i, y_i, z_i, 1]$ with RHS R

Why correct: The source presents sphere fitting as the 3D analogue of circle fitting and adds the z -coordinate in exactly the same squared-distance expansion pattern.

Memorize / understand: Circle pattern + one dimension: add $-2z$ on the left and $-z^2$ on the right.

5. Trick-wording decoder and common distractors

Goal: Train recognition: many MCQs do not test a new concept; they disguise a known formula by moving a transpose, changing a sign, or switching homogeneous/inhomogeneous language.

If the question says...	Translate it mentally to...
"Residual" / "error vector" / "difference between sides"	All point to $Ax=b$ in the inhomogeneous least-squares section.
"Normal equation"	$A^T A x = A^T b$. Do not confuse "normal" here with curve normals from gradients.
"Gradient of a quadratic form"	General A: $(A^T + A)x$. Symmetric matrix: $2Ax$.
"Curve normal" / "implicit normal"	Differentiate $J(x,y)$ with respect to x and y ; the gradient is normal to the level set.
"Homogeneous"	Right-hand side is zero: $Mz=0$. Zero solution always exists.
"Inhomogeneous"	Right-hand side is not identically zero: $Ax=b$. Circle fitting is written this way in the source.
"Overdetermined"	More equations than unknowns ($m>n$), usually requiring a least-squares estimate.
"Pseudoinverse"	In the tall full-column-rank case, connect it to $(A^T A)^{-1}A^T$.
"Circle design matrix"	Left row is $[1, -2x, -2y]$; right side is negative squared coordinates.
"Ellipse design matrix"	Row is $[x^2, xy, y^2, x, y, 1]$; do not mix it with the circle row.
"Scale ambiguity"	In a homogeneous implicit equation, multiplying all coefficients by a nonzero scalar describes the same zero set.
"Discriminant"	Ellipse condition in these notes: $B^2 - 4AC < 0$; scale-fixed constraint is set to -1 .

Dimension check as a weapon: If options look similar, check dimensions. For A in $R^{(m \times n)}$, $A^T A$ is $n \times n$, $A^T b$ is $n \times 1$, and x is $n \times 1$; this instantly eliminates many wrong answers.

Sign check as a weapon: For circle fitting, expand $(x-x_0)^2$: the cross term is negative. Moving $x_i^2 + y_i^2$ to the right makes the RHS negative.

6. Numerical micro-drills

Method: Cover the answer line with your hand, solve quickly, then compare. These target the mistakes that appeared during our practice: least-squares arithmetic and radius recovery.

1. $A=[1,1,1]^T$, $b=[2,4,6]^T$. Least-squares x ?

- A. 4
- B. 2
- C. 6
- D. 12

Answer: A. x is the average here: $(2+4+6)/3=4$.

2. $A=[1,2]^T$, $b=[3,7]^T$. Least-squares x ?

- A. $17/5$
- B. $10/3$
- C. 5
- D. 2

Answer: A. $A^T A=5$ and $A^T b=17$, so $x=17/5$.

3. If $c=-12$, $x_0=3$, $y_0=4$, find R .

- A. $\sqrt{37}$
- B. 7
- C. 5
- D. $\sqrt{12}$

Answer: A. $R^2=9+16-(-12)=37$.

4. Circle center $(2,1)$, point $(5,5)$. grad J ?

- A. $[6,8]^T$
- B. $[3,4]^T$
- C. $[-6,-8]^T$
- D. $[5,5]^T$

Answer: A. $2[(5-2),(5-1)]^T=[6,8]^T$.

5. $J=3x^2+2xy+5y^2$. grad J ?

- A. $[6x+2y, 2x+10y]^T$
- B. $[3x+2y, 2x+5y]^T$
- C. $[6x, 10y]^T$
- D. $[6x+2x, 2y+10y]^T$

Answer: A. Differentiate componentwise; xy contributes y to d/dx and x to d/dy , including coefficient 2.

6. $A=[[1,2],[3,4]]$, $x=[1,2]^T$. grad of $x^T A x$?

- A. $[12,21]^T$
- B. $[10,22]^T$
- C. $[13,19]^T$
- D. $[14,20]^T$

Answer: A. $(A^T+A)x = [[2,5],[5,8]][1,2]^T=[12,21]^T$.

7. For one circle point $(2,-1)$, what is the design row?

- A. $[1,-4,2]$
- B. $[1,4,-2]$
- C. $[4,1,1]$
- D. $[1,-2,-1]$

Answer: A. $[1,-2x,-2y]=[1,-4,2]$.

8. For one circle point $(2,-1)$, what is RHS?

- A. -5
- B. 5
- C. -2
- D. 3

Answer: A. $-x^2-y^2=-4-1=-5$.

9. If B is symmetric, $\text{grad}(x^T B x)$ at x is?

- A. $2Bx$
- B. $(B^T+B)x$ only, cannot simplify
- C. Bx
- D. $B^T Bx$

Answer: A. Symmetry gives $B^T=B$, so the sum is $2B$.

10. If M has full column rank, exact solutions of $Mz=0$?

- A. Only $z=0$
- B. Infinitely many nonzero z
- C. None
- D. Only $z=1$

Answer: A. Full column rank means $\text{null}(M)=\{0\}$.

7. Four 5-question mini-mocks

Use these like the real quiz: For each set, first answer without looking at the key. Then review only the questions you missed or guessed.

Mock A - core recognition

1. For $m > n$ in $Ax=b$, what is the standard objective?

- A. $\min \|Ax-b\|_2^2$
- B. $\min \|A^T x-b\|_2^2$
- C. $\min \|x-b\|_2^2$
- D. $\min \|Ax\|_2^2$

2. Which is the normal equation?

- A. $A^T A x = A^T b$
- B. $AA^T x = b$
- C. $A^T x = b$
- D. $Ax = A^T b$

3. General $\text{grad}(x^T A x)$?

- A. $(A^T + A)x$
- B. $2Ax$ always
- C. $A^T A x$
- D. Ax

4. Circle fitting RHS for (x_i, y_i) ?

- A. $-x_i^2 - y_i^2$
- B. $x_i^2 + y_i^2$
- C. $-2x_i - 2y_i$
- D. R^2

5. Circle normal for $J=0$?

- A. $[2(x-x_0), 2(y-y_0)]^T$
- B. $[x_0, y_0]^T$
- C. $[y, -x]^T$
- D. $[R, R]^T$

Key: 1A, 2A, 3A, 4A, 5A | Review explanations in the earlier archetype cards.

Mock B - transpose/sign traps

1. If $B=B^T$, $\text{grad}(x^T Bx)=?$

- A. $2Bx$
- B. Bx
- C. $2B^T Bx$
- D. $B^T x$

2. Circle design row?

- A. $[1, -2x_i, -2y_i]$
- B. $[x_i^2, y_i^2, 1]$
- C. $[1, x_i, y_i]$
- D. $[-x_i^2, -y_i^2, 1]$

3. If $c=x_0^2+y_0^2-R^2$, $R=?$

- A. $\sqrt{x_0^2+y_0^2-c}$
- B. $\sqrt{c-x_0^2-y_0^2}$
- C. $x_0^2+y_0^2-c$
- D. $\sqrt{x_0+y_0-c}$

4. Ellipse condition?

- A. $B^2-4AC<0$
- B. >0
- C. $=0$
- D. $A+C=0$

5. Homogeneous system always has?

- A. $z=0$
- B. no solution
- C. $z=1$
- D. a unique nonzero solution

Key: 1A, 2A, 3A, 4A, 5A | Review explanations in the earlier archetype cards.

Mock C - reasoning

1. Why impose $\mathbf{z}^T \mathbf{K} \mathbf{z} = -1$ in ellipse fitting?

- A. Fix scale and exclude $\mathbf{z} = 0$
- B. Make $\mathbf{M}$ square
- C. Force center at origin
- D. Make norm negative

2. Which row belongs to ellipse $\mathbf{M}$?

- A. $[x^2, xy, y^2, x, y, 1]$
- B. $[1, -2x, -2y]$
- C. $[x, y, 1]$
- D. $[x^2, y^2, 1]$

3. Gradient of $\mathbf{J} = \mathbf{A}\mathbf{x}^2 + \mathbf{B}\mathbf{x}\mathbf{y} + \mathbf{C}\mathbf{y}^2 + \mathbf{D}\mathbf{x} + \mathbf{E}\mathbf{y} + \mathbf{F}$?

- A. $[2\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} + \mathbf{D}, \mathbf{B}\mathbf{x} + 2\mathbf{C}\mathbf{y} + \mathbf{E}]^T$
- B. $[2\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} + \mathbf{D}, 2\mathbf{C}\mathbf{x} + \mathbf{B}\mathbf{y} + \mathbf{E}]^T$
- C. $[\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} + \mathbf{D}, \mathbf{B}\mathbf{x} + \mathbf{C}\mathbf{y} + \mathbf{E}]^T$
- D. $[2\mathbf{A}\mathbf{x}, 2\mathbf{C}\mathbf{y}]^T$

4. What does full column rank imply for $\mathbf{M}\mathbf{z} = 0$?

- A. Only $\mathbf{z} = 0$
- B. Infinite nonzero $\mathbf{z}$
- C. No solution
- D. $\mathbf{M}$ must be square

5. Why is gradient normal to $\mathbf{J} = 0$?

- A. It is perpendicular to level sets
- B. It is tangent
- C. It has unit norm
- D. It points to origin

Key: 1A, 2A, 3A, 4A, 5A | Review explanations in the earlier archetype cards.

Mock D - mixed calculation

1. $A=[1,2]^T, b=[3,7]^T$. x is?

- A. $17/5$
- B. 5
- C. $10/3$
- D. 2

2. $c=-12, x_0=3, y_0=4$. R ?

- A. $\sqrt{37}$
- B. 7
- C. 5
- D. $\sqrt{12}$

3. At circle center (2,1), point (5,5), grad J ?

- A. $[6,8]^T$
- B. $[3,4]^T$
- C. $[-6,-8]^T$
- D. $[5,5]^T$

4. $\text{grad}(x^T x)$?

- A. $2x$
- B. x
- C. 2
- D. x^T

5. Ellipse Lagrangian gives?

- A. $M^T Mz = -\lambda Kz$
- B. $Mz = \lambda z$
- C. $Kz = 0$
- D. $M^T z = -\lambda z$

Key: 1A, 2A, 3A, 4A, 5A | Review explanations in the earlier archetype cards.

8. Source notes, apparent typos, and final checklist

Important: source typo #1 (Matrix p.4): The displayed expansion of $\|Ax-b\|^2$ appears to print the middle term with x where b should be. The next derivative and the normal equation correctly use b : $\text{grad } J = 2A^T A x - 2A^T b$ and $A^T A x = A^T b$. For quiz preparation, memorize the derivative and normal equation.

Important: source typo #2 (Circle/Ellipse p.4): The PDF appears to print the second component of the ellipse gradient as $2Cx+Bx+E$. Mathematically, d/dy of Cy^2 is $2Cy$, so the correct second component is $Bx+2Cy+E$. If the quiz reproduces the PDF verbatim, be aware of the discrepancy; if it asks mathematics, use the correct derivative.

Possible sphere typo: The sphere equation in the source appears to contain a coordinate-label typo in one displayed term. The matrix formulation directly below follows the expected 3D pattern with x , y , and z center coordinates.

Source map

Source location	What to study there
Matrix (1).pdf pp.1-2	Notation, vector derivative definition, derivative of $x^T x$, scalar product derivative.
Matrix (1).pdf p.3	Quadratic-form derivative, symmetric special case, start of inhomogeneous overdetermined systems.
Matrix (1).pdf p.4	Least-squares objective, gradient, normal equation, closed-form solution, start of line fitting.
Matrix (1).pdf p.5	Line/plane fitting case studies (secondary to stated quiz topics).
Circle Ellipse Fitting.pdf p.1	Circle equation, linearized matrix system, inverse/pseudoinverse statement.
Circle Ellipse Fitting.pdf p.2	Sphere extension, general conic/ellipse equation, ellipse condition, $Mz=0$ formulation.
Circle Ellipse Fitting.pdf p.3	Scale constraint, K matrix, Lagrangian, generalized eigenproblem, circle normal.
Circle Ellipse Fitting.pdf p.4	Ellipse normal/gradient; apparent typo flagged above.

Final 10-minute checklist

- ☐ Can I write $A^T A x = A^T b$ without looking?
- ☐ Can I write $x=(A^T A)^{-1}A^T b$ and state when it is valid?
- ☐ Can I distinguish $\text{grad}(x^T A x)=(A^T+A)x$ from the symmetric shortcut $2Ax$?
- ☐ Can I reconstruct circle row $[1,-2x,-2y]$ and RHS $-x^2-y^2$?
- ☐ Can I explain $c=x_0^2+y_0^2-R^2$ and recover R ?
- ☐ Can I write the circle normal $[2(x-x_0), 2(y-y_0)]^T$?
- ☐ Can I explain why a gradient is normal to a level curve?
- ☐ Can I explain homogeneous $Mz=0$, zero solution, and scale ambiguity?
- ☐ Can I state the ellipse condition $B^2-4AC<0$ and row $[x^2, xy, y^2, x, y, 1]$?
- ☐ Can I spot a transpose/sign/dimension trap quickly?

Night-before strategy: Do not spend your last hour rereading every derivation. Write the emergency formulas from memory, do Mock D, then revisit only the cards tied to mistakes or guesses.